

Class 12 Biology: High-Yield Diagram & Mark Breakdown

Chapter 4: Principles of Inheritance and Variation (NCERT Page-wise Guide)

Instructions for Students: Practice genetic crosses with complete Punnett squares, parental genotypes, and clear genotypic/phenotypic ratios. Ensure pedigree symbols are standardized.

Fig No.	Cross / Diagram Title	NCERT	Marks	Priority	Key Components / Ratios to Practice
Fig 4.7	Dihybrid Cross (Seed Shape & Colour)	Pg 63	5 M	Top Priority	RRYY × rryy → 16-box Punnett square. Phenotypic ratio: 9:3:3:1
Fig 4.4	Monohybrid Cross (Punnett Square)	Pg 57	3 M	Top Priority	TT × tt → F ₁ (Tt). F ₂ Phenotypic (3:1), Genotypic ratio (1:2:1)
Fig 4.6	Incomplete Dominance (Snapdragon)	Pg 60	3 M	Top Priority	RR (Red) × rr (White) → Rr (Pink). F ₂ Phenotypic = Genotypic (1:2:1)
Fig 4.5	Test Cross (Dominant Phenotype)	Pg 59	2 – 3 M	Top Priority	Crossing unknown violet flower with homozygous recessive (ww); 1:1 if heterozygous
Fig 4.15	Sickle-Cell Anaemia (HbA vs HbS)	Pg 74	3 M	Top Priority	Point mutation: GAG → GUG codon; Glu → Val at 6th position of β-globin chain
Fig 4.13	Pedigree Analysis Symbols	Pg 72	2 M	Top Priority	Male (□), Female (○), Affected (filled), Mating, Consanguineous mating
Fig 4.14	Pedigree Chart Interpretation	Pg 73	3 M	High Priority	(a) Autosomal Dominant (Myotonic dystrophy), (b) Autosomal Recessive (Sickle-cell)
Fig 4.13 (b)	Sex Determination in Honey Bee	Pg 71	2 – 3 M	High Priority	Haplodiploidy: Female 2n=32 (meiosis), Male n=16 (mitosis/p partenogenesis)
Fig 4.12	Sex Determination Systems	Pg 70	2 – 3 M	High Priority	(a, b) Male heterogamety: XY/XX (Human, Drosophila), (c) Female heterogamety: ZW/ZZ (Birds)
Fig 4.11	Morgan's Linkage & Recombination	Pg 68	3 – 5 M	High Priority	Cross A (y-w: 1.3% recomb - tight linkage), Cross B (w-m: 37.2% recomb - loose linkage)

EXAM STRATEGY & GUIDELINES

- **5-Mark Dihybrid Cross:** Always show Gametes (RY, Ry, rY, ry), the full 16-box Punnett grid, and write both Phenotypic (9:3:3:1) and Genotypic ratios clearly.
- **Sickle-Cell Molecular Detail:** Show exact template/mRNA codons (GAG → GUG) and amino acid replacement at position 6 (Glutamic acid → Valine).